

Project 3 — Windows Network Connection & Endpoint Investigation

Controlled Windows Endpoint SOC Investigation

1. Executive Summary

This investigation examined outbound network activity from a controlled Windows 11 SOC lab endpoint and attempted to correlate network activity with endpoint process telemetry.

A controlled HTTPS connection to www.microsoft.com succeeded from the lab endpoint (10.0.2.15) to 95.100.165.206 over TCP/443. Windows native TCP telemetry subsequently provided process-attributable HTTPS connections for Microsoft/Windows components including StartMenuExperienceHost.exe, svchost.exe hosting WpnService, and Microsoft Edge WebView2.

Sysmon was confirmed to be running with a valid configuration containing a NetworkConnect filter. However, sysmon -c repeatedly reported Network connection: disabled, and no Sysmon Event ID 3 records were generated after controlled HTTPS activity. This established a telemetry gap rather than evidence of an absent network connection.

The observed process paths, publishers and execution context were consistent with legitimate Windows/Microsoft components. The activity was assessed as Benign / Controlled Lab Activity — Low Suspicion.

2. Investigation Objective

Determine whether outbound network activity can be observed on the endpoint, identify the responsible process where possible, associate the process with a user/session and process-creation context, and assess whether the activity appears benign or suspicious.

A secondary objective was to validate whether Sysmon Event ID 3 (Network Connection) was actually providing the expected telemetry.

3. Environment & Telemetry

Endpoint: Windows 11 SOC laboratory VM

Endpoint IP: 10.0.2.15

Sysmon service: Sysmon

Sysmon driver: SysmonDrv

Sysmon operational log: Microsoft-Windows-Sysmon/Operational

Relevant Sysmon events: Event ID 1 (Process Create), Event ID 3 (NetworkConnect)

Native Windows telemetry used:

- Test-NetConnection
- Get-NetTCPConnection
- Get-Process
- Sysmon Event ID 1

Sysmon status was verified with both Sysmon and SysmonDrv running.

4. Controlled Network Test

Command:

Test-NetConnection www.microsoft.com -Port 443

Observed result:

ComputerName: www.microsoft.com

RemoteAddress: 95.100.165.206

RemotePort: 443

InterfaceAlias: Ethernet

SourceAddress: 10.0.2.15

TcpTestSucceeded: True

This confirms that the endpoint successfully established a TCP connection to the tested HTTPS destination. The later Get-NetTCPConnection snapshot did not contain 95.100.165.206, so the later persistent connections were not attributed to this specific short-lived test.

5. Sysmon Event ID 3 Validation

The Sysmon configuration contained a NetworkConnect include filter. The configuration was validated and loaded successfully. However, sysmon -c reported: Network connection: disabled.

A fresh Test-NetConnection test succeeded, but querying the Sysmon operational log for Event ID 3 returned no matching events.

Finding: Sysmon Event ID 3 network telemetry was unavailable on this endpoint despite the NetworkConnect filter being present. This is recorded as a telemetry limitation and not as evidence that the network connection failed.

6. Native Network Telemetry

Get-NetTCPConnection -RemotePort 443 identified the following process-attributable HTTPS connections during the current investigation.

7. Process Correlation

Sysmon Event ID 1 records were correlated to the observed PIDs. PID reuse was explicitly considered: PID 4012 had previously belonged to SecurityHealthHost.exe on the prior day, but the current PID 4012 was StartMenuExperienceHost.exe. Therefore, PID alone was not treated as a persistent identity; current process creation time and process context were used.

Current process creation evidence:

- PID 6640: msedgewebview2.exe, Microsoft Edge WebView2, user SOCAlyst\SOCAlyst2, LogonId 0x91145, TerminalSessionId 1, IntegrityLevel AppContainer. Parent was another msedgewebview2.exe process.
- PID 4012: StartMenuExperienceHost.exe, Microsoft Windows component, user SOCAlyst\SOCAlyst2, LogonId 0x91145, TerminalSessionId 1, IntegrityLevel Medium. Parent was svchost.exe -k DcomLaunch -p running as SYSTEM.
- PID 3344: svchost.exe -k netsvcs -p -s WpnService, Microsoft Windows component, user SYSTEM, LogonId 0x3E7, TerminalSessionId 0, IntegrityLevel System. Parent was services.exe.

8. Network-to-Process Evidence

Current Get-NetTCPConnection snapshot:

- 10.0.2.15:55059 → 52.110.3.7:443, Established, PID 4012
- 10.0.2.15:55058 → 52.110.3.52:443, Established, PID 4012
- 10.0.2.15:53756 → 172.187.86.74:443, Established, PID 3344
- 10.0.2.15:52561 → 2.17.113.112:443, CloseWait, PID 6640

These connections provide current process-to-network attribution through native Windows telemetry. They should not be represented as the specific connection generated by the earlier Test-NetConnection command, because that test targeted 95.100.165.206.

9. Assessment

Overall assessment: Benign / Controlled Lab Activity — Low Suspicion.

Reasons supporting the assessment:

- The controlled HTTPS test succeeded without any indication of malicious behavior.

- The process paths are standard Windows or Microsoft application locations.
- The identified binaries are Microsoft components.
- The process execution contexts and parent processes are consistent with normal Windows application/service operation.
- No persistence, privilege escalation, credential access, or suspicious command-line behavior was identified in the collected evidence.

Caution: This assessment is limited to the available endpoint telemetry. Lack of Sysmon Event ID 3 should not be interpreted as proof that no network activity occurred.

10. Telemetry Gap & SOC Lesson

The principal finding is a telemetry gap: Windows native TCP telemetry exposed active HTTPS connections and owning process IDs, while Sysmon Event ID 3 was not available even though the NetworkConnect filter was configured.

SOC lesson: detection engineering must validate the complete telemetry path — configuration, feature state, event generation, log ingestion and process correlation — rather than assuming that a configured data source is producing the expected events.

11. MITRE ATT&CK; Mapping

Primary technique:

- T1049 — System Network Connections Discovery

Supporting endpoint discovery context:

- T1057 — Process Discovery

These mappings describe the investigative techniques demonstrated by the lab rather than asserting adversary activity.

12. Evidence & Reproducibility

Key commands used:

- Get-Service Sysmon,SysmonDrv
- sysmon -c
- Test-NetConnection www.microsoft.com -Port 443
- Get-NetTCPConnection -RemotePort 443
- Get-Process -Id 4012,3344,6640
- Get-WinEvent ... Id=1

Important reproducibility note: network socket state is dynamic. Short-lived connections may disappear before collection and may appear as TIME_WAIT with OwningProcess 0. PID reuse can also occur, so process creation time and ProcessGuid should be preferred over PID alone when available.

6.1 Current Network Connections

PID	Process	Destination	State	User / Context
4012	StartMenuExperienceHost.exe	52.110.3.7:443	Established	SOCAnalyst\SOCAnalyst2
4012	StartMenuExperienceHost.exe	52.110.3.52:443	Established	SOCAnalyst\SOCAnalyst2
3344	svchost.exe (WpnService)	172.187.86.74:443	Established	SYSTEM
6640	msedgewebview2.exe	2.17.113.112:443	CloseWait	SOCAnalyst\SOCAnalyst2